

Early warning for thermal runaway in lithium-ion batteries during various charging rates: Insights from expansion force analysis

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ABSTRACT

Thermal runaway introduces a significant challenge in the widespread application of lithium-ion batteries, necessitating advanced early-warning technologies to ensure safety, particularly during charging. Only monitoring the temperature and voltage limit the performance of diagnostic algorithms. The expansion behavior of batteries, which is linked to their operating status, offers a vital indicator of safety. In this study, the evolution of multidimensional signals during overcharging experiments at different current rates is comprehensively investigated. The result shows that the abnormal expansion force can be detected at temperatures as low as 35.4 °C, which achieves an early warning signal 11 min earlier than the onset of battery thermal runaway. The effect of charging rate on battery safety is comprehensively analyzed, showing that the time interval between the warning signal of the expansion force and temperature increases steadily from 151 s to 682 s as the charging rate decreases. However, the charging rate hardly affects the stage of charge boundary of venting, which is around ± 118 %. These insights are crucial for understanding early warning mechanisms in overcharged batteries, offering valuable guidance for enhancing the safety of electric vehicles and energy storage systems.

Nomenclature

Abbreviations	
CO	Carbon monoxide
CO ₂	Carbon dioxide
CR	Charging rate
CT	Computed Tomography
ESS	Energy storage systems
EV	Electric vehicle
SC	Short circuit
LiFePO ₄	Lithium iron phosphate
LIB	Lithium-ion battery
SEI	Solid electrolyte interface
SOC	State of charge
TR	Thermal runaway
VOCs	Volatile organic compounds

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Symbols	
dF/dt	Expansion force change rate (N/s)
$(dF/dt)_{Max}$	Maximum expansion force rise rate (N/s)
F	Expansion force (N)
F_V	Force of venting (N)
T	Battery bottom-surface temperature (°C)
T_{Max}	Maximum temperature (°C)
T_{abn}	Temperature of expansion force becoming abnormal (°C)
T_{SC}	Temperature of internal short-circuit (°C)
T_{TR}	Temperature of thermal runaway (°C)
T_V	Temperature of venting (°C)
t_{abn}	Time for expansion force to become abnormal (s)
t_{ISC}	Time for the occurrence of internal short-circuit (s)
t_{TR}	Time for the occurrence of thermal runaway (s)
t_V	Time of venting (s)

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Δt_1	Time interval of dF/dt advance expansion force
Δt_2	Time interval of dF/dt advance voltage
Δt_3	Time interval of dF/dt advance temperature
U_v	Voltage of venting (V)
Subscrip	
abn	abnormal
v	venting

1. Introduction

The development of renewable energy sources, electric vehicles (EVs), and energy storage systems (ESSs) is essential for addressing the global energy crisis (Shahzad et al., 2021; Tan et al., 2023; Li et al., 2023). Lithium-ion batteries (LIBs) have emerged as a dominant power source owing to their improved performance and decreased production (Abu et al., 2023; Che et al., 2023; Osmani et al., 2023). However, numerous accidents involving thermal runaway (TR) in LIBs have highlighted potential safety concerns and hindered their application (Wang et al., 2022a; Hong et al., 2024; Zhou et al., 2024). In particular, thousands of LIB cells are stacked in EVs or ESSs, and the poor battery consistency and fault of the battery management system can further increase the risks associated with overcharge-induced TR (Tian et al., 2024; Wu et al., 2023; Zhou et al., 2022). In particular, fast charging significant amounts of flammable gases and heat are released (Zhu et al., 2023; Jia et al., 2024; Wang et al., 2024a). Once TR is triggered, failure propagation occurs among the numerous cells and a significant amount of energy is released rapidly, thus resulting in a disastrous fire or explosion (Wang et al., 2022b; He et al., 2023) and further causing pollutant emissions and resource waste (Wei et al., 2023). Therefore, early safety-warning technology is an important strategy for ensuring battery charging safety (Wu et al., 2014; Chen et al., 2023a; Jia et al., 2023a).

Several studies have contributed to the development of early-warning methods for ensuring battery-charging safety (Kong et al., 2023; Yang et al., 2023a; Huang et al., 2023). Zhang et al. (2023) classified the LiFePO₄ overcharge process into four stages and proposed a three-level warning strategy based on the voltage turning point, temperature, and appearance of various gases. This strategy can detect overcharges and mitigate the risk of TR. Jin et al. (2020) developed a sensitive method via hydrogen (H₂) capture to detect lithium dendrite growth in LIBs. They discovered that H₂ was detected earlier than that of other gases during overcharging experiments. Charging stoppages upon H₂ detection can effectively reduce the risk of TR. Wang et al. (2020) identified five discernible voltage inflection points during overcharging at varying charging rates (CRs). These inflection points were associated with processes such as lithium deposition, electrolyte oxidation, reactions between the electrolyte and deposited lithium, separator shrinkage, and short circuits (SCs). Monitoring these voltage inflection points allows one to identify potential TR incidents. Jiang et al. (2019) analyzed the duration of voltage plateau during overcharging at various CRs to predict the occurrence of TR. They discovered that the voltage behavior is indicative of potential TR incidents. Su et al. (2021) introduced an early-warning method for ESSs using venting acoustic signals, which offered advantages such as rapid implementation, high sensitivity, and low cost; furthermore, it achieved a recognition accuracy of 92.31 %. Song et al. (2022) monitored the air pressure in a confined space to issue a warning signal before battery TR. They demonstrated that a warning signal can be issued, on average, 473 s before TR, thus providing an ample early-warning period. However, in practice, due to the complexity of multiple physical fields in the process of overcharge-induced TR for LIBs, the signals above present problems such as poor stability and long response time and cannot be used to monitor the operating status and failure mechanism of batteries in a timely and accurate manner.

The expansion force is considered a potential warning signal for battery failure owing to its rapid response, high reliability, and low cost (Koch et al., 2018; Li et al., 2024a), which renders it more advantageous compared with other signals such as temperature, voltage, venting acoustics, and gas composition. Chen et al. (2023b) proposed force-capacity equations during battery TR. Furthermore, they introduced a force magnification-based structure named, ‘no TR propagation’ to achieve active and passive safety for LIBs. Lv et al. (2023) investigated variations in the expansion force during overcharging, over-discharging, and heating-plate-induced TR. They proposed an approach for early anomaly detection using expansion-force anomalies, which enabled the identification of different types of failures based on the observed behavior of expansion-force changes. Zhou et al. (2023) conducted a thorough investigation into the TR phenomenon in LIBs under real-world conditions, delving into the intricate relationship between TR behaviors and variables such as preload force and charging rate. They discovered that TR triggered by overcharging is closely linked to expansion force signals, and is significantly influenced by both the preload force and the charging rate. It was found that higher charging rates escalate the risk of experiencing TR, whereas a greater preload force can improve the resistance for LIBs to overcharging but at the cost of intensifying the severity of TR. Jia et al. (2023b) explored the gas venting dynamics during TR incidents triggered by both overcharging and overheating. They noted that overcharging leads to substantial gas production, causing safety vents to activate sooner compared to overheating scenarios. During overheating, it’s primarily the vaporization of electrolytes that prompts safety venting in LIBs. The volumes of gas released during TR incidents initiated by overcharging and overheating were measured at 62.1 L and 101.3 L, respectively, highlighting that gases released in overheating scenarios are non-toxic. Nevertheless, the intricate patterns of multi-dimensional signal evolution, including expansion force across varying charging rates, remain uncharted territory. However, in the most typical overcharge-induced TR scenario, the evolution mechanism of the expansion force signal in the entire process remains ambiguous, particularly its comprehensive comparison with those of other signals. Furthermore, the effects of the CR on the time sequence and safety boundary of the force signal, as well as the effectiveness of the early-warning system utilizing the expansion force have not been quantified. Early safety-warning evaluation approaches based on expansion force signals are limited, particularly for scenarios involving fast charging.

This study focuses on analyzing variations in the expansion force, voltage, temperature, and gas signals during the overcharging of LIBs, specifically within a CR range of 1 C–6 C. Nondestructive characterization methods, such as ultrasonic and computed tomography (CT) scanning experiments, have been employed to quantitatively evaluate the gas-generation behavior and assess the internal structural damage of LIBs. This study presents a significant advancement in providing early overcharge warnings for LiFePO₄ batteries, particularly crucial for scenarios involving high-rate fast charging, and marks a notable contribution to the fields of battery safety and early warning systems as follows.

- (1) Multidimensional signal evolution mechanisms are comprehensively revealed under overcharged abuse at different CRs, especially for expansion force.
- (2) The effect of CR on battery-safety boundary parameters, such as the expansion force characteristics, gas-generation behavior, and internal structure, is qualitatively and quantitatively analyzed.
- (3) The expansion force is utilized as an early-overcharging warning signal with experimental validation at the cell level, particularly for charging at high CRs.

2. Experiment

This section provides information about the experimental methods in

our study.

2.1. Battery sample

Table 1 presents a comprehensive overview of the prismatic batteries used in this study. The battery is composed of graphite and lithium iron phosphate (LiFePO₄) and has a capacity of 40 Ah. The battery measures 148 mm (length) × 27.5 mm (thickness) × 130 mm (height), and the nominal voltage, maximum cut-off voltage, and minimum cut-off voltage of the battery are 3.2, 3.65, and 2.0 V, respectively. The weight of the batteries is (1065 ± 10) g and the initial stage of charge (SOC) is set to 0 %.

2.2. Experimental settings

An experimental system was established to acquire multidimensional signals, such as the expansion force, gas compositions, voltage, and surface temperature of the battery during overcharging. The constant acquisition frequency of the signals is 1.0 Hz. Fig. 1 (a) depicts the connection diagram of the experimental devices, which includes a test fixture, a battery cycler, an abuse test chamber, a data-acquisition system, a computer, and a camera. As shown in the real photos of the experimental apparatus in Fig. 1 (b), the test fixture comprised three steel plates and eight M10 bolts. Both sides of the test battery sample were similarly clamped by the side mica boards and fixture plates, to mitigate heat leakage during battery failure. In addition, an external force sensor was nested on the grooved mica plate with matching dimensions and then clamped by two stainless steel plates, to measure the external expansion force of the LIBs. Five gas sensors, including volatile organic compounds (VOCs), H₂, carbon monoxide (CO), carbon dioxide (CO₂), and CH₄ were arranged on the top bracket of the abuse test chamber, to monitor the gas compositions. Two K-type thermocouples were positioned in the central region of the battery surface to monitor the surface temperature. After applying a preload of 1000 N to the battery through screw rotation based on the references in (Chen et al., 2023a; Li et al., 2024a), the batteries were subjected to overcharge abuse experiments. The batteries were overcharged in the constant-current mode under a CR of 1–6 C.

In addition, to determine the effect of the CR on the gas-generation boundaries and internal battery structures, nondestructive ultrasonic scanning and CT scanning were performed for quantitative analysis, and the experimental diagrams are shown in Fig. 1 (c) and (d), respectively. For the ultrasonic scanning procedure, the battery specimen is immersed in silicone oil to facilitate C scanning. The settings for the probe include a frequency of 2.25 MHz, a focal length of 25.4 mm, and an emission voltage set to 200 V. The ultrasonic scanning is conducted with a step size of 0.2 mm and achieves an imaging resolution of 0.3 mm. Further specifics regarding the ultrasonic scanning methodology are detailed in reference (Li et al., 2024b). Moreover, the operating voltage and current during CT scanning are 240 kV and 340 μA, respectively.

Table 1
Detailed specifications of tested LiFePO₄ battery.

Parameter	Specification
Cathode	LiFePO ₄
Anode	Graphite
Nominal capacity (Ah)	40.0 ± 1.0
Dimensions (mm)	148 × 27.5 × 130
Nominal voltage (V)	3.2
Maximum cut-off voltage (V)	3.65
Minimum cut-off voltage (V)	2.0
Mass (g)	1065 ± 10
Energy density (Wh/kg)	121.9
Initial SOC (%)	0

2.3. Error analysis

The experimental errors involved in this study are mainly caused by the accuracy error of the experimental instrument, the inconsistency in the manufacturing of each battery sample, and the difficulty of completely consistent manual operation of the experimental device. The error analysis of the accuracy of the experimental instrument is summarized in Table 2. In addition, the sampling frequency of the data is constant at 1.0 Hz, and the side reaction after the battery thermal runaway is triggered is extremely violent. Therefore, there could be certain errors in parameters such as maximum temperature and venting force. To minimize the occasionalities in experimental outcomes and enhance the reliability of the research findings, each CR under electrical abuse conditions was subjected to repeated trials.

3. 3. results and discussions

3.1. Multidimensional signal evolution during overcharging

Fig. 2 shows the expansion force, voltage, and temperature characteristics observed during the overcharging test. Several key moments and stages were defined based on these signals. The moment characterized by a rapid increase in the rate of the expansion force, which reached 5 N/s within 3 s, is denoted as t_{abn} , because when the expansion force exhibits abnormal characteristics its rise rate increases more sharply. The phenomenon of safety-valve venting is denoted by t_v . This occurs when the pressure inside the battery exceeds a certain threshold, and the safety valve is activated to release excess gas. The point at which the voltage drop rate reaches its maximum is denoted as t_{sc} , whereas the point at which the temperature rise rate reaches its peak is denoted as t_{tr} . Based on these key moments, three characteristic peaks of the expansion force are identified, which correspond to venting, SC, and TR. The entire overcharging process is classified into five stages: Stage I begins from a fully charged battery and continues until t_{abn} , i.e. when the abnormal expansion force begins. Stage II spans the period from t_{abn} until safety valve venting (t_v) occurs. Stage III encompasses the period between the onset of safety valve venting and the occurrence of an SC (t_{sc}). Stage IV spans from the occurrence of an SC to the battery reaching its maximum temperature rise (t_{tr}). Finally, stage V extends from the TR and continues until the end of the test.

Stage I: After the batteries reach 100 % SOC, no discernible abnormalities are observed in the expansion force, voltage, and temperature signals. This can be attributed to the surplus capacity of the battery electrode material (Wang et al., 2021). The surplus capacity allows for the incorporation of lithium ions into the anode material, which is the primary cause of the expansion force observed during this stage. The following reactions occur during this stage (Oh et al., 2014):



Stage II: During this overcharging stage, the voltage and temperature gradually increase as the battery continues to be overcharged. Additionally, the expansion force increases significantly, which can also be found in (Lv et al., 2023; Wang et al., 2024b). This trend owes primarily to the initiation of side reactions within the battery, which generates irreversible gases (Yang et al., 2023b). As these side reactions occur, gas generation within the battery increases. Once the internal gas reaches a specific threshold, the safety valve is activated, thus resulting in the release of white smoke. The battery behavior varies with the CR. At 1 C and 2 C, the battery voltage does not exhibit significant changes prior to the venting of the safety valve. However, as the CR reaches 3 C, the voltage increases significantly before the venting of the safety valve. This may be because the polarization increases with the battery charging rate increasing. As the amount of gas production is related to the overcharge capacity, a higher charging voltage is required to achieve the same gas

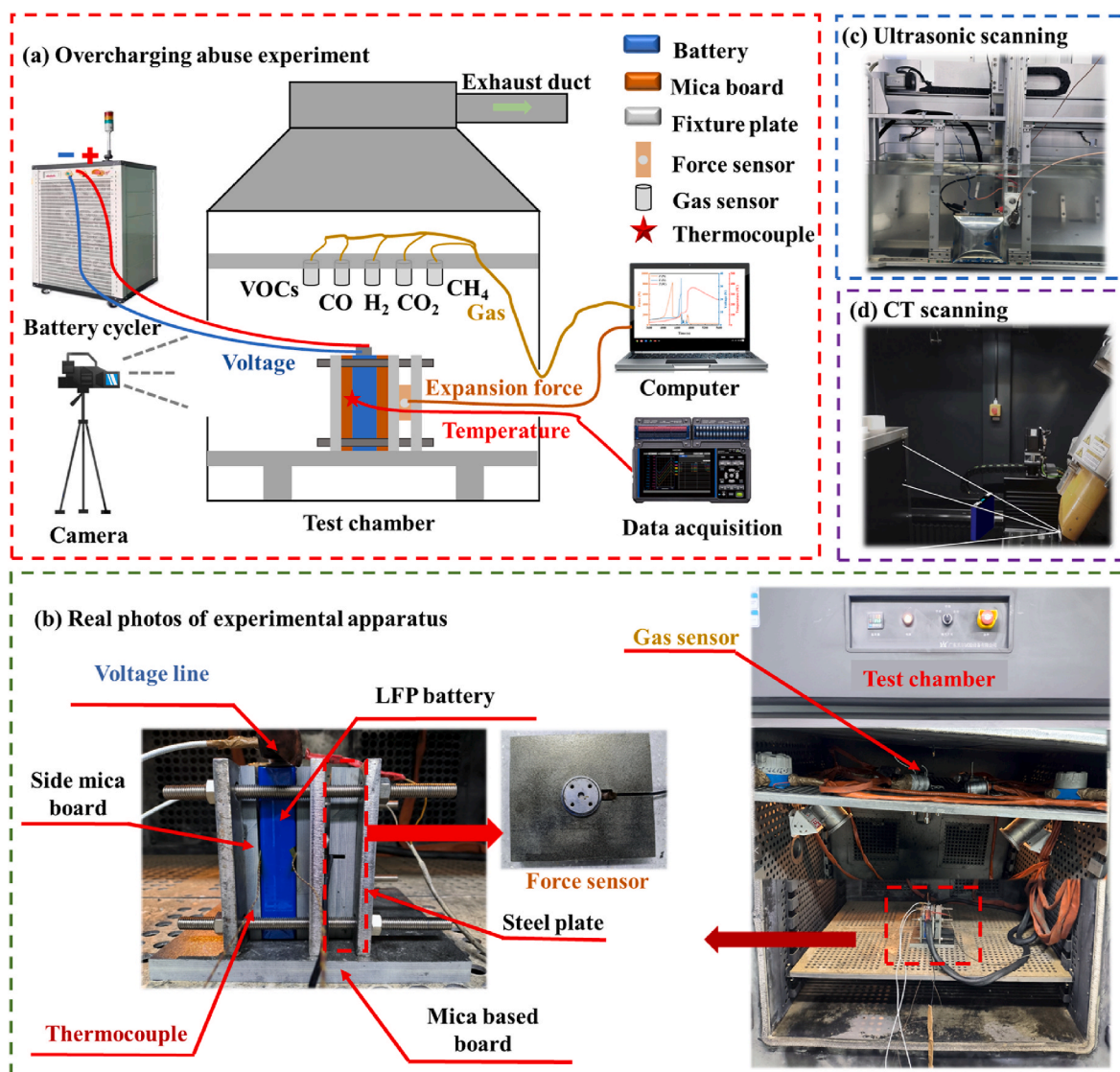


Fig. 1. Experimental details. (a) Schematic illustration of overcharge experimental setup, diagram of test fixture, and arrangement of force sensor and thermocouples; (b) real photos of experimental apparatus; (c) ultrasonic scanning experiment; and (d) CT scanning experiment.

Table 2
Specifications of experimental devices.

Device	Model	Range	Accuracy
Battery cycler	NEWARE CT-8008 -100V 400A-NTFA	0.25–100 V 0.5–400 A	±0.1 %
Data acquisition	HIOKI LR 8450	–200–2000 °C	±0.1 °C
Force sensor	Z 673	0–10000 N	±0.3 %
Thermocouple	OMEGA GG-K-30	0–1250 °C	±1.1 °C

production (overcharge capacity). Therefore, from Fig. 2, we can see intuitively that the battery voltage shifts before the safety valve releases gas with the charging rate increasing.

Stage III: In this stage, several significant changes occur within the battery. After the safety valve is vented in Stage II, the expansion force decreases rapidly. However, the voltage continues to increase anomalously, and the battery temperature increases gradually. The anomalous increase in voltage triggers a series of side reactions within the battery (Wang et al., 2020, 2024a), and these reactions result in electrolyte decomposition and increase the expansion force. At a certain point, the battery voltage increases abruptly and then

decreases significantly. This indicates the occurrence of a battery SC. Subsequently, the expansion force decreases, and the battery temperature stabilizes.

Stage IV: As the separator within the battery collapses slightly, the voltage continues to decrease gradually. In addition, the battery temperatures at 1 C and 2 C decrease slightly, which can be attributed to heat dissipation from the battery to the ambient. However, once the separator is melted completely, a significant event occurs within the battery. The voltage decreases to 0 V, thus indicating a complete loss of electrical potential. This abrupt decrease in voltage results in a significant increase in temperature within the battery. An increase in the temperature further induces other gas-generation side reactions, which can result in the release of a significant amount of smoke from the battery. Consequently, the external expansion force signal exhibits a third characteristic peak.

Stage V: After triggering TR in the battery, the temperature begins to decrease after reaching its characteristic peak. The expansion force and voltage stabilize gradually as time progresses.

3.2. Changing rates of the multidimensional signal during overcharging

The change rates of the expansion force, voltage, and temperature

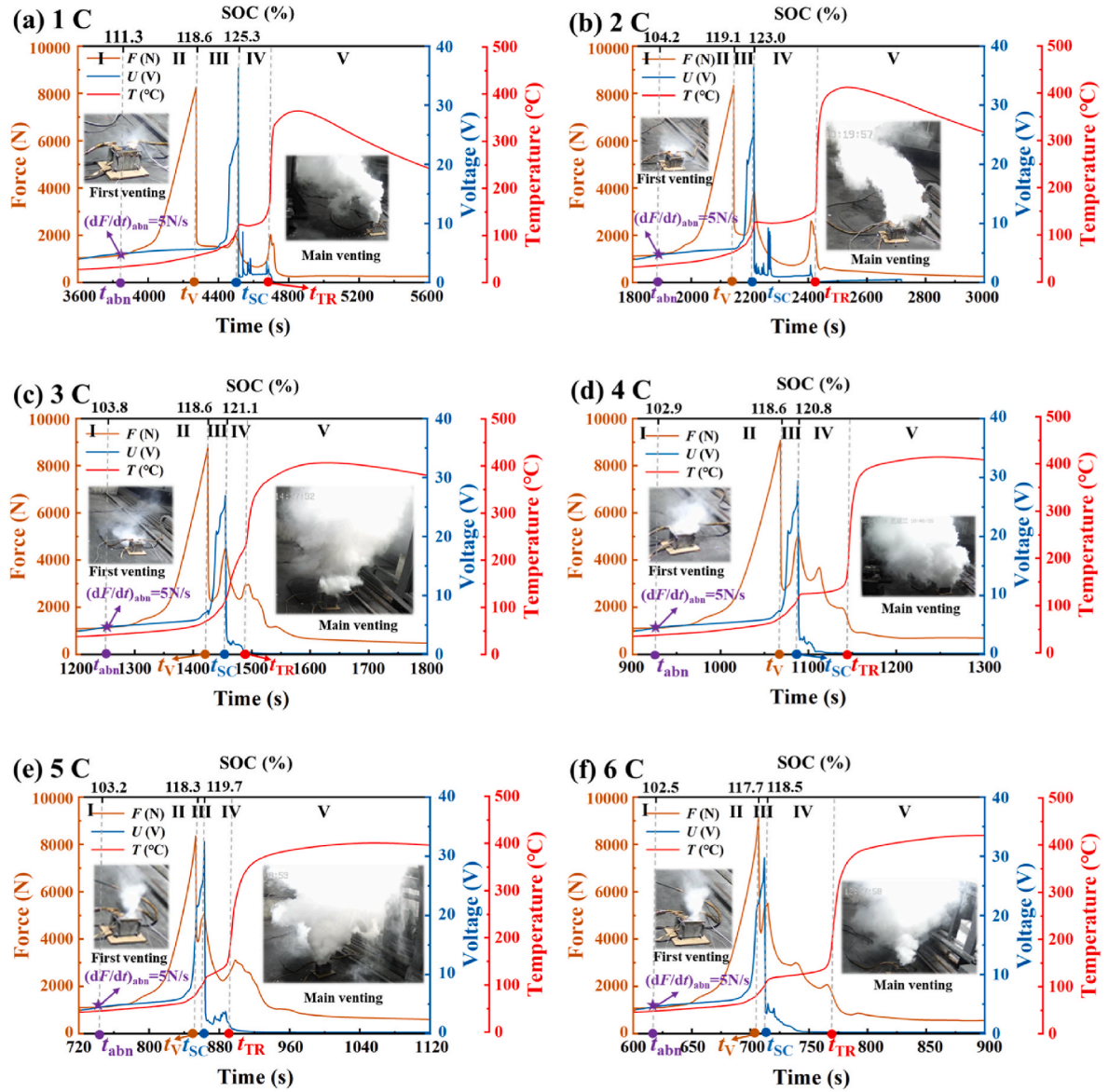


Fig. 2. Expansion force, voltage, and temperature evolution during overcharging at different charging rates. (a) 1 C; (b) 2 C; (c) 3 C; (d) 4 C; (e) 5 C; (f) 6 C.

during the overcharging test are shown in Fig. 3, which includes the five stages mentioned above.

Stage I: The change rates of the expansion force, voltage, and temperature did not indicate any clear abnormalities.

Stage II: At the onset of this stage, the expansion-force rise rate exhibited anomalous variations before the voltage and temperature rise rate, the growth in the force is consistent with that reported in reference (Yang et al., 2023b). This suggests that changes in the expansion force occurred earlier and may serve as an early indicator of abnormalities during overcharging. In stage II, the voltage and temperature change rates exhibited abnormal behavior, particularly when the CR exceeded 2 C.

Meanwhile, the expansion-force rise rate showed various characteristics. A plateau was indicated in the expansion-force rise rate, as shown in Fig. 4 (a), (b), and (c). However, when the CR reached 4 C, no clear plateau was observed in the expansion-force rise rate, as shown in Fig. 4 (d), (e), and (f). The absence of a plateau suggests the occurrence of other side reactions before the venting of the safety valve, particularly

at higher CRs. The fitted form of the battery expansion force is expressed as

$$y = a + b * x \quad (2)$$

The fitted equations are expressed as shown in Eq. (3)–(8). The R-squared values of all the fitting curves exceeded 0.98. The gradual increase in the slope of the fitted curve with the CR was significant. This finding suggests a positive correlation between the CR and the intensity of the side reactions occurring within the battery during overcharging.

$$F = -119584.87 + 29.91 t \quad (3)$$

$$F = -109948.52 + 55.10 t \quad (4)$$

$$F = -137255.45 + 102.32 t \quad (5)$$

$$F = -131710.62 + 131.36 t \quad (6)$$

$$F = -141406.12 + 174.93 t \quad (7)$$

$$F = -153650.92 + 228.90 t \quad (8)$$

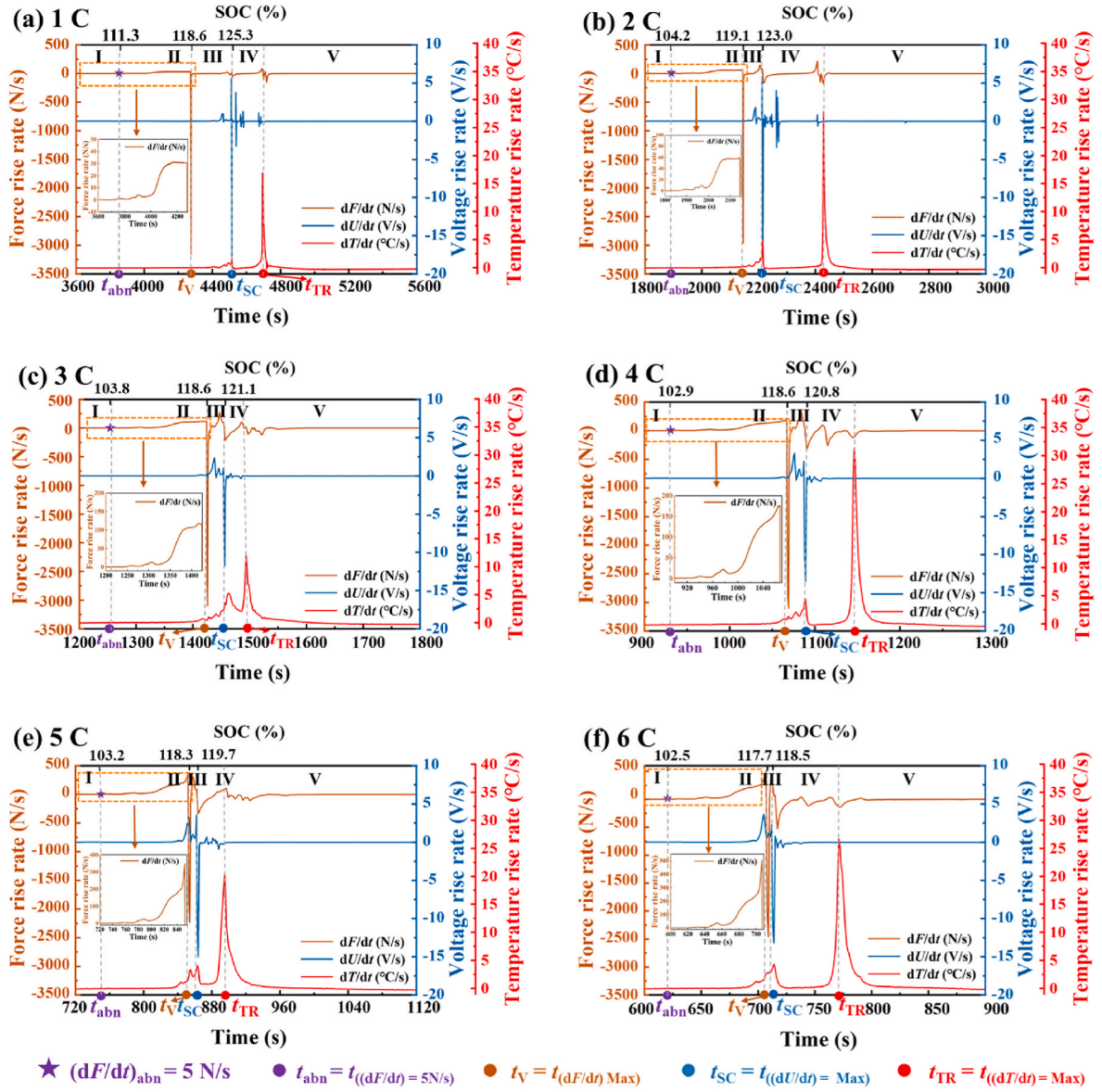


Fig. 3. Change rates of expansion force, voltage, and temperature during overcharging at different charging rates. (a) 1 C; (b) 2 C; (c) 3 C; (d) 4 C; (e) 5 C; (f) 6 C.

where F refers to the expansion force during the change plateau period and t is the duration of this period.

Stage III: After the safety valve was vented, the voltage change rates fluctuated significantly, which can be attributed to the occurrence of SC within the battery. In addition to the fluctuating voltage change rates, the expansion force and temperature rise rates increased progressively. The rates of expansion force and temperature change began to decrease as the rates of voltage changed significantly. This decrease in the expansion force and temperature change rates was caused by the occurrence of SC within the battery, which resulted in a rapid release of energy.

Stage IV: This stage represents the transitional region between SC and TR, which is caused by the low-temperature rise rates. Simultaneously, the change rates of the expansion force and voltage fluctuated. TR was triggered until heat accumulated to a certain extent within the battery when the temperature change rate increased rapidly, followed by a rapid decrease.

Stage V: In the later stage of the overcharging process, the change rates of the expansion force, voltage, and temperature progressively approached zero.

3.3. SOC boundary of expansion force, voltage, and temperature

Fig. 5 illustrates the variations in the expansion force, voltage, and temperature with the SOC at different CRs. Fig. 5 (a-1) and (a-2) depict the correlation between the expansion force and its change rate with the SOC. In general, the SOC boundaries corresponding to multiple signal abnormal moments are as follows: the expansion force signal is the lowest, followed by the voltage signal, and the temperature signal is the highest. The expansion force and its rise rate at 6 C charging were greater than that at 1 C. This difference is attributable to the fact that lithium deposition is facilitated by overcharging at a higher CR (Arai et al., 2015). The expansion-force rise rate exhibited three characteristic peaks. The first peak occurred when the SOC reached approximately 105 %. The second characteristic peak was observed at approximately 108 %–110 % SOC. Subsequently, the expansion-force rise rate first decreased and then increased, which was accompanied by the

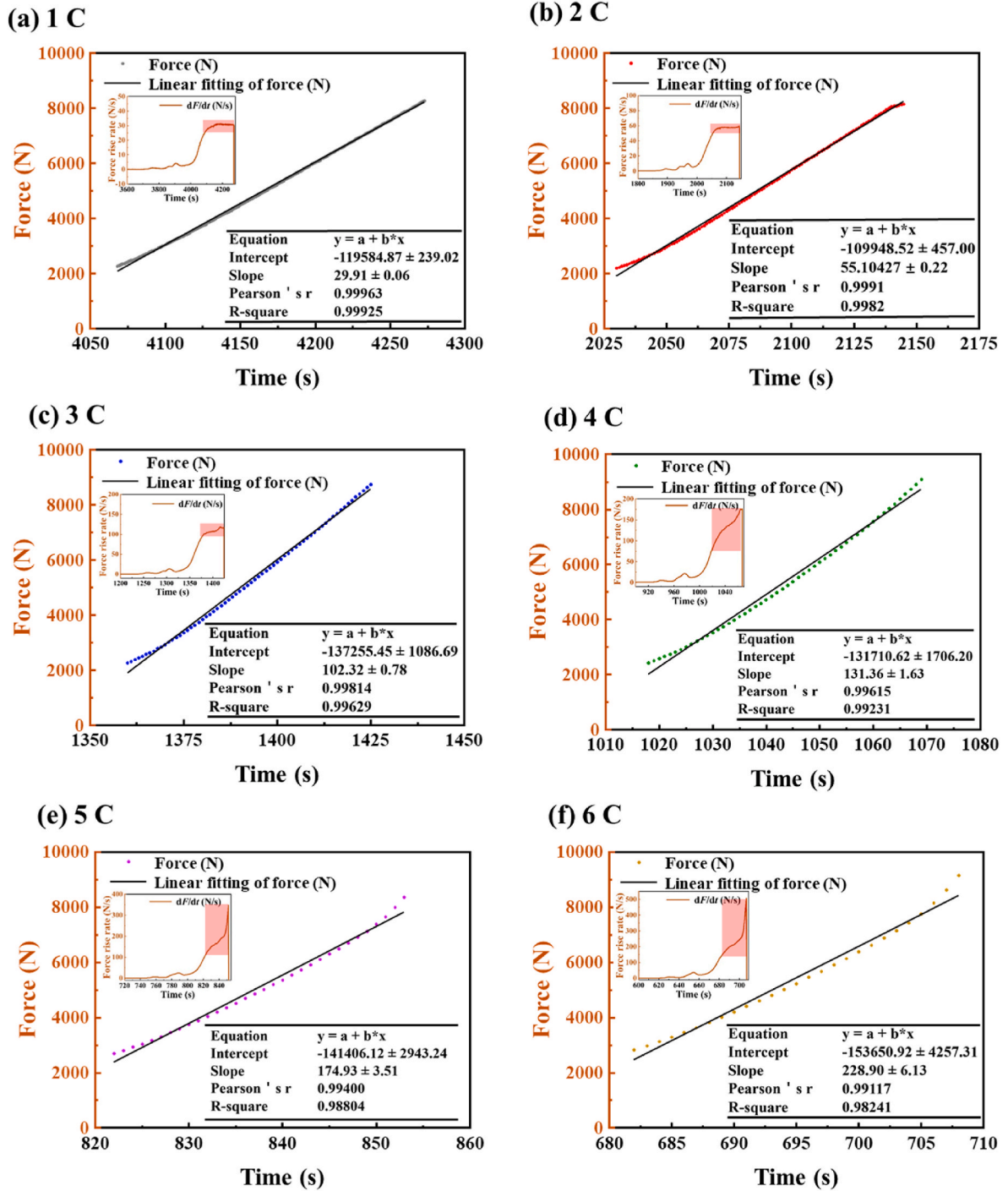


Fig. 4. Linear fitting results during the plateau period of the expansion force rise rate. (a) 1 C; (b) 2 C; (c) 3 C; (d) 4 C; (e) 5 C; (f) 6 C.

occurrence of the third characteristic peak. However, repeatable experimental results show that the corresponding SOC of the battery venting has an error of $\pm 2\%$, which may be due to the inconsistency in battery safety valve manufacturing and is within the error range. Therefore, in general, the CR minimally affected the SOC boundary when the safety valve vented, i.e. it remained at approximately $\pm 118\%$ SOC. Fig. 5 (b-1) and (b-2) show the correlation between voltage and the SOC, as well as the relationship between the voltage rise rate and SOC, respectively. When the voltage began to increase abnormally, the corresponding SOC boundary decreased as the CR increased, which is consistent with reference in (Xu et al., 2021). Fig. 5 (b-2) indicates the

presence of two distinct peaks in the voltage change rate. The first peak value increased gradually with the CR. Quantitatively, the voltage of the battery subjected to overcharging at rates of 1 and 6 C showed clear anomalies at 123 % and 115 % SOC, respectively. When the CR was between 1 and 2 C, the second peak value was greater than the first peak value. However, when the CR reached 3 C, the opposite was observed. Fig. 5 (c-1) shows the correlation between temperature and the SOC. When the SOC was 120 %–135 %, the temperature exhibited a plateau. Fig. 5 (c-2) demonstrates the relationship between the temperature change rate and SOC, whose general trend agrees well with Fig. 4 (b) from reference (Wang et al., 2024a). The temperature change rate

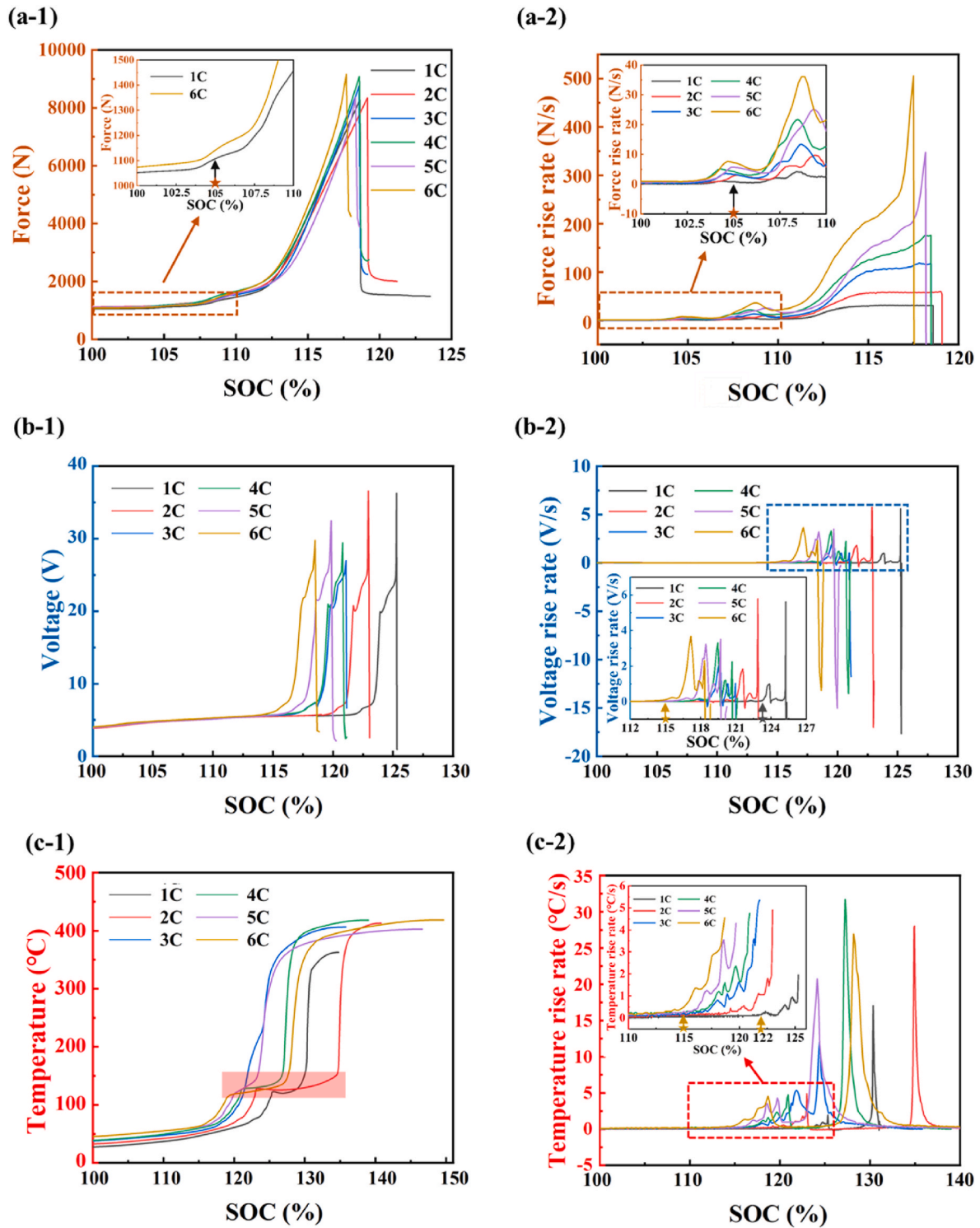


Fig. 5. Anomalous SOC boundaries for expansion force, voltage, and temperature. (a-1) Expansion force vs. SOC; (a-2) Expansion force rise rate vs. SOC; (b-1) Voltage vs. SOC; (b-2) Voltage rise rate vs. SOC; (c-1) Temperature vs. SOC; (c-2) Temperature rise rate vs. SOC.

exhibited two characteristic peaks. As the CR increased, the SOC boundary of the first characteristic peak advanced. Specifically, in the cases of 1 C and 6 C overcharging, the battery temperature increased significantly at SOC of 122 % and 115 %, respectively. In addition, the higher the battery charging rate, the faster the lithium-ion migration rate from the cathode to the anode, the charging current increases, and the battery heat generation increases (Wang et al., 2024a). Therefore, the battery will show a temperature rise earlier.

3.4. Coupling mechanisms of multidimensional signals

Fig. 6 shows the interconnections among the various signals. Fig. 6 (a) and (b) display the expansion force and its change rate, respectively, as a function of temperature. The expansion force increased unnaturally at 35 °C during overcharging at 1 C. Generally, as the CR increased, the temperature at which the expansion force exhibited anomalous behavior increased progressively, thus suggesting that the expansion force was

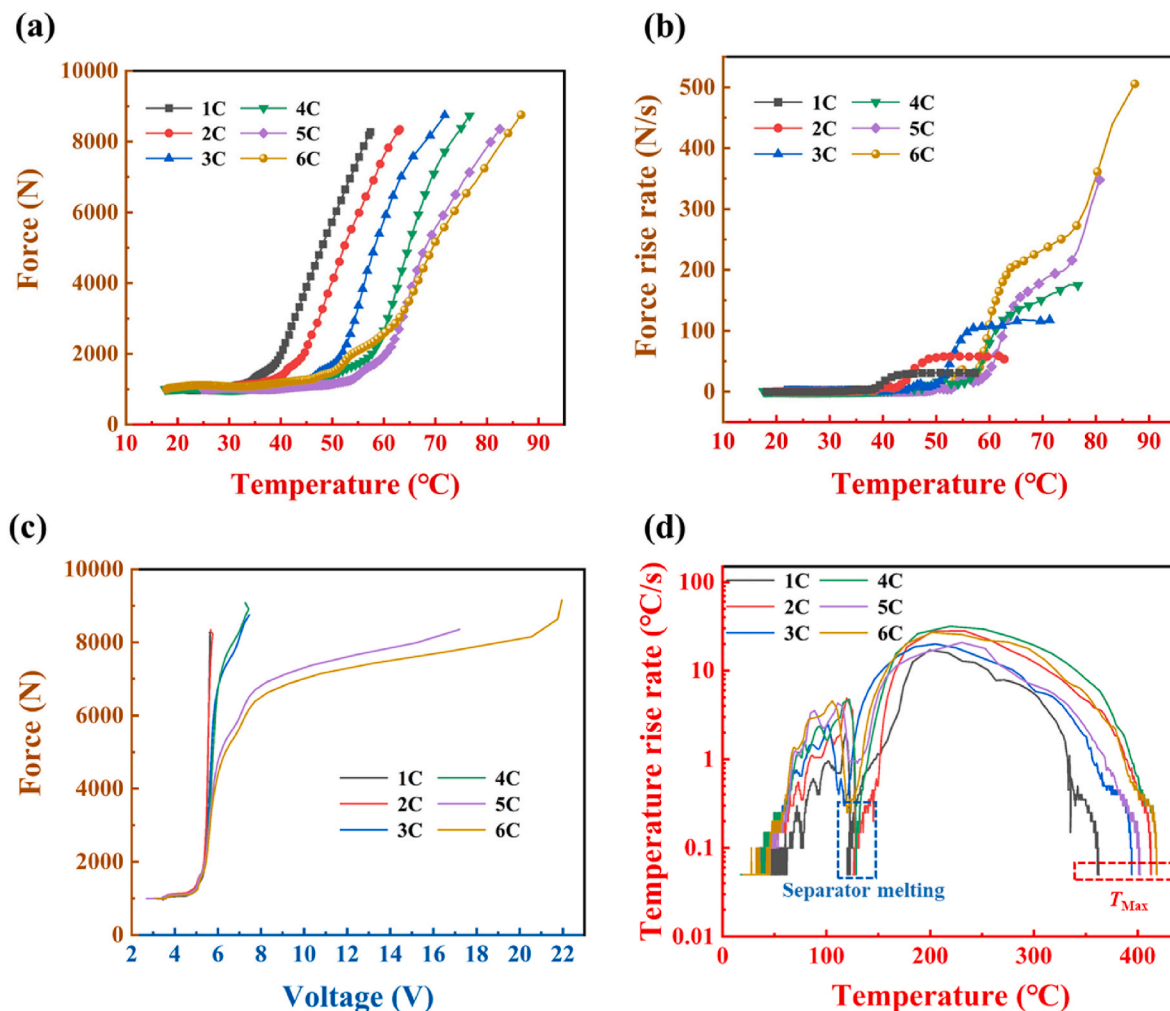


Fig. 6. Coupling mechanisms of multidimensional signals under different charging rates: (a) Force vs. temperature; (b) force rise rate vs. temperature; (c) force vs. voltage; (d) temperature rise rate vs. temperature.

more responsive than the temperature under the overcharged state. Nevertheless, the curve of 6 C is not in line with this trend in Fig. 6 (a) may be due to the inconsistency of the LIBs or the experimental errors in manually operated devices. Fig. 6 (b) further demonstrates that the expansion-force rise rate did not change significantly when the battery temperature was lower than 40 °C. However, as the overcharge rate increased, the change characteristics of the expansion-force rise rate became more pronounced. In particular, charging at 5 C and 6 C significantly increased the expansion-force rise rate. This can be attributed to that the higher charging rate causes quicker growth in the overcharge capacity, and the faster gas production rate, hence, the higher expansion force rise rate. In addition, a higher charging rate accelerates heat generation and elevates the temperature rise. This part of heat further aggravates the side reaction of gas production (Wang et al., 2020, 2024a). Consequently, the higher the charging rate, the greater the expansion force rise rate.

Fig. 6 (c) shows the correlation between the expansion force and voltage. When the voltage reached 5 V, all batteries exhibited a significant increase in the expansion force. This is primarily due to the decomposition of the electrolyte inside the battery, which triggered a series of gas-generating side reactions (Chen et al., 2024). When the battery was overcharged at 5 C or 6 C, the voltage increased significantly and rapidly once the expansion force reached 4000 N. The higher the CR, the higher the sensitivity of the voltage signal, thus causing it to change significantly before the safety valve was vented. This suggests

that the expansion force signal is more appropriate for safety warnings in the early stage of the temperature rise triggered by the high-rate overcharging of the battery. Fig. 6 (d) exhibits the relationship between the temperature rise rate and temperature. Before an SC occurred within the battery, the temperature rise rate increased with the CR. When the battery experienced an SC, the temperature rise rate decreased. When the battery experienced TR, the temperature rise rate increased rapidly and reached its peak at approximately 200 °C. Subsequently, the change rate of temperature decreased gradually.

3.5. Effect of charging rate on gas-generation behaviors

A series of experiments were conducted in this study to determine the SOC boundary of gas generation during the overcharging process and to evaluate the effect of the CR on the gas generation behavior. Scanning was employed to detect internal gas generation within the batteries. Fig. 7 displays the ultrasonic transmission scanning images obtained during the overcharging process. The batteries underwent overcharging at rates of 1 C and 6 C through ultrasonic transmission scanning. Ultrasonic scanning in C-scan mode necessitates a specific duration to complete, and it was observed that ultrasonic scan images exhibit significant differences once the SOC surpasses 105 %. Consequently, scan images for representative scenarios at various SOCs, namely 100 %, 101 %, 103 %, and 105 %, are presented for both 1 C and 6 C charging rates independently. The scanning results show that the battery initially

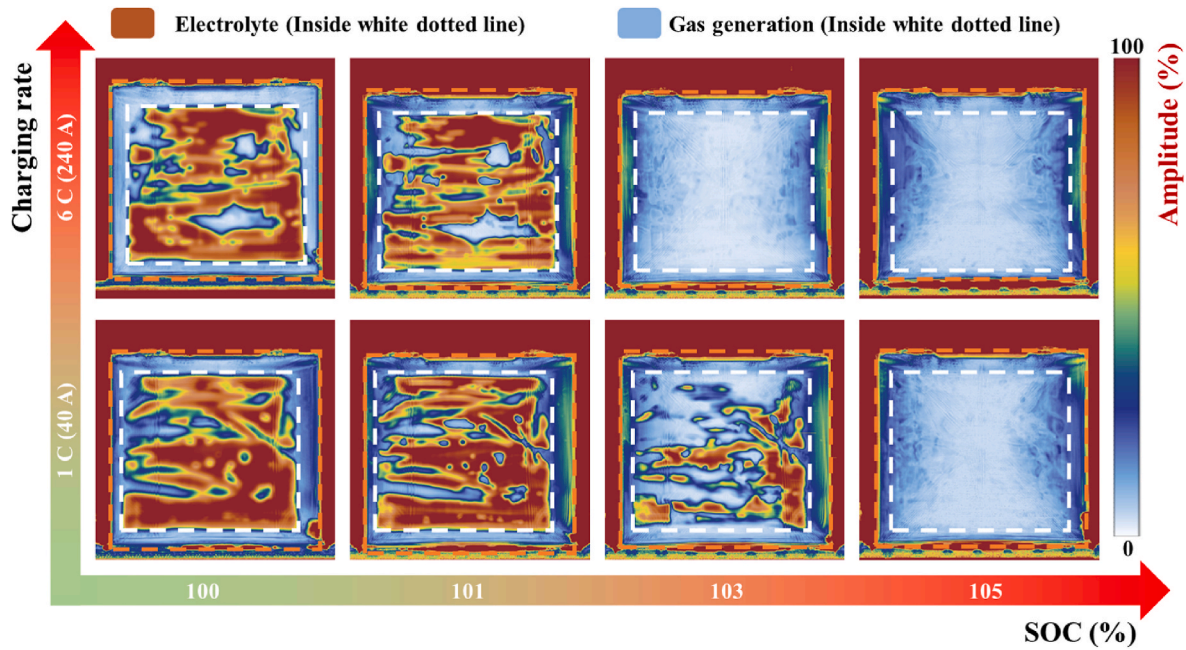


Fig. 7. SOC boundaries of gas generation at 1 C and 6 C charging rates based on ultrasonic scanning. Note: The area enclosed by the orange dotted is the battery, the area enclosed by the white dotted line is the jelly rolls, the area between the orange and white dotted lines is the air gap between the core and the shell, and the dark red area beyond the orange dashed line is the silicone oil medium. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

exhibited manufacturing faults that resulted in trace amounts of gas within the battery before the experiments. As the battery was overcharged up to 103 % SOC, the battery overcharged at 1 C and began to generate a certain amount of gas. However, the battery charged at 6 C

was completely filled with gas in this status. The battery overcharged at 1 C and was not completely filled with gas until its SOC reached 105 %. Therefore, one can conclude that the SOC boundary at which the battery is completely filled with gas decreases as the CR increases during

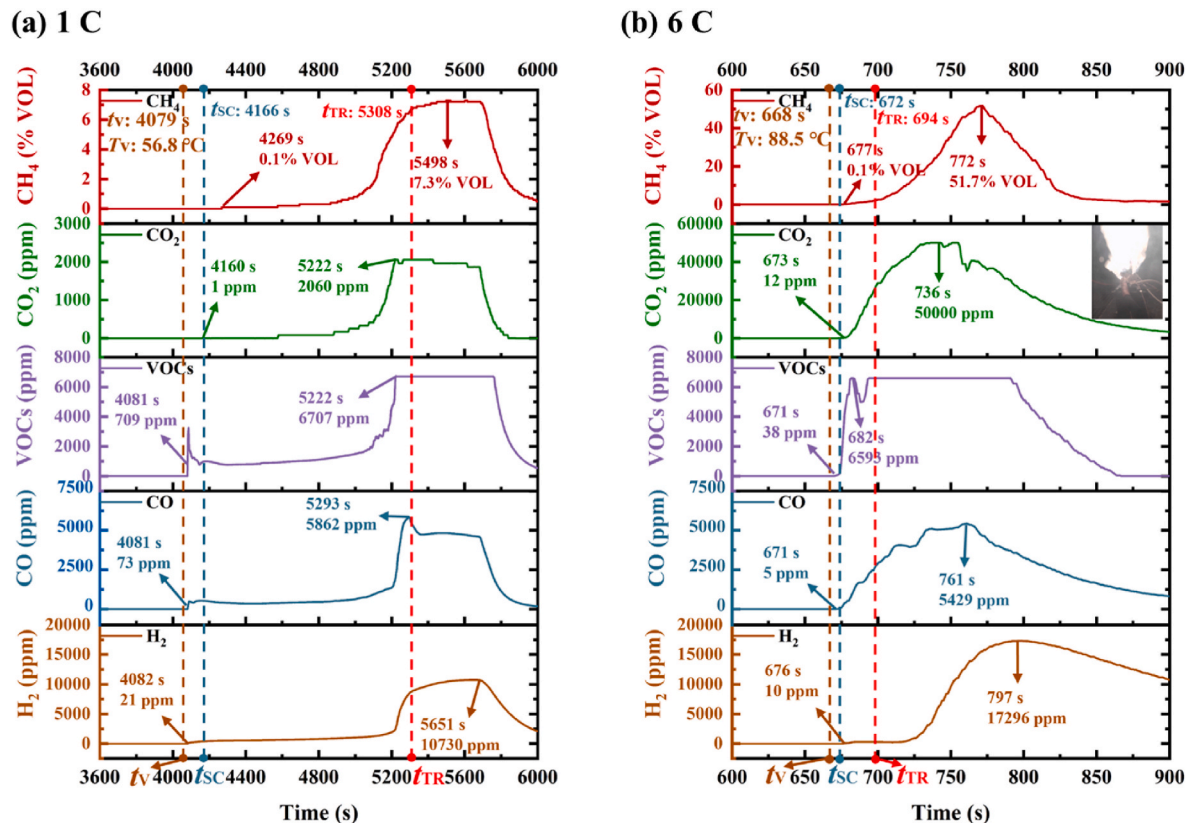


Fig. 8. Variation in gas concentration during overcharging at charging rates of 1 C and 6 C.

overcharging.

During battery TR induced by overcharging, several gases, including H_2 , VOCs, CO, CO_2 , and CH_4 are generated (Pham et al., 2020; Ren et al., 2019; Huang et al., 2021). Fig. 8 (a) and (b) depict the gas evolution during the overcharging process at 1 C and 6 C, respectively. At a CR of 1 C, the battery detected only VOCs, CO, and H_2 after the safety valve was venting for 2–3 s, thus suggesting that the safety valve vented primarily because of these three gases. However, CH_4 and CO_2 were not detected until the battery was subjected to an SC. When the temperature reached a certain threshold, a significant number of gases were rapidly generated. In addition, the amounts of CO_2 and CH_4 generated at 6 C were significantly higher than those generated at 1 C. The significantly higher peaks of CO_2 were likely due to the battery catching fire after the safety valve vented. The significantly higher peaks of CH_4 can be attributed to the increased deposition of Li, which reacted with the electrolyte at high overcharge rates (Liu et al., 2022).

3.6. Effect of charging rate on battery internal structure

CT, which is a nondestructive characterization method, has been widely employed to evaluate the internal structure of batteries (Weng et al., 2021; Finegan et al.; Rahe et al., 2019). CT scanning tests were conducted on battery wreckages subjected to overcharging at rates of 1 C and 6 C, and the results are presented in Fig. 9. The findings from the CT scanning revealed that the thickness of the horizontal section in the XY plane was larger than that of the vertical section in the YZ plane. This indicates that the internal structure of the battery wreckage was disturbed by gas generation during overcharging. Additionally, the thickness of the battery wreckage overcharged at 6 C was greater than that of the battery wreckage overcharged at 1 C. This suggests that batteries subjected to overcharging at higher CRs exert a more pronounced effect on their internal structures, which can be attributed to two factors. At high CRs, the deposition of Li at the anode increases (Ye et al., 2016), which can result in the formation of dendrites and cause

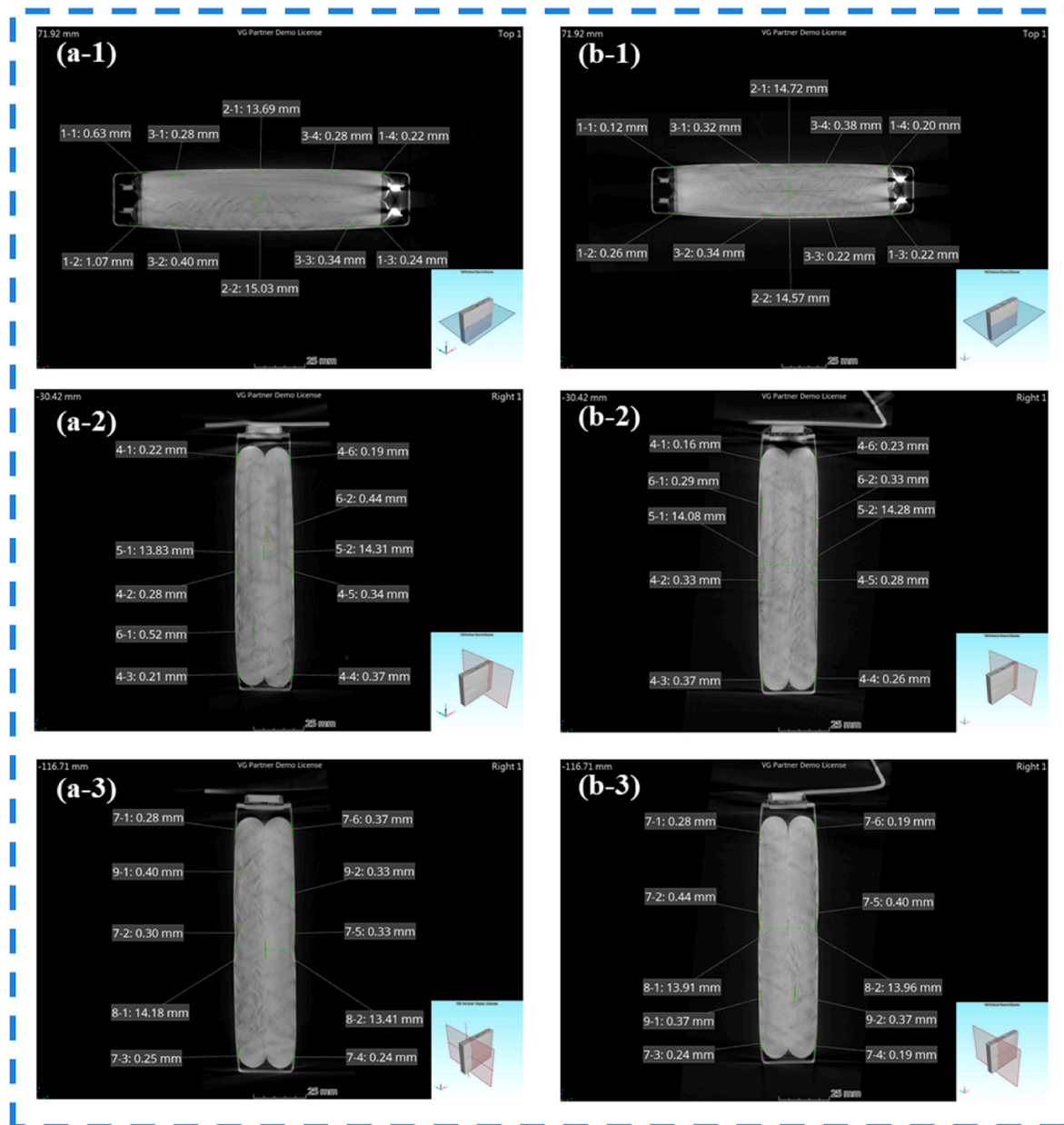


Fig. 9. CT scanning images of batteries. (a) Battery wreckage after overcharge test at 1 C and (b) battery wreckage after overcharge test at 6 C; (–1) horizontal section in the XY plane, (–2) vertical section in YZ plane of cathode, and (–3) orthogonal slice in YZ plane of anode.

structural damage within the battery. Second, more violent gas generation (Hendricks et al., 2015) during overcharging at high CRs can cause more severe damage to the electrode materials and the overall battery structure.

3.7. Effect of charging rate on battery-safety boundary parameters

Various characterization parameters for different CRs were quantitatively evaluated. As shown in Fig. 10 (a), the expansion force of venting (F_V) was not significantly affected by the CR. The fluctuation in F_V is attributable to the manufacturing process of the safety valve instead of to the CR. Fig. 10 (b) shows that the venting voltage (U_V) increased progressively with the CR. This suggests that a higher CR results in greater polarization. Hence, a more pronounced voltage anomaly when the safety valve vents. Fig. 10 (c) shows that the temperature corresponding to a force rise rate of 5 N/s was 35.4 °C–48.1 °C, which increased with the CR. This suggests that early warning based on the

expansion force can be successfully executed within the normal temperature range below 50 °C. The venting temperature (T_V) ranged from 56.8 °C to 88.6 °C and exhibited a positive correlation with the CR. This is primarily because the battery generated more Joule heat when charging at higher rates. These findings suggest that the expansion force signal may be more valuable for early warning than the temperature signal before the safety valve vents. Based on the temperature at which the voltage drop rate reached the maximum (T_{SC}), the T_{SC} at CRs of 1–4 C was approximately 120 °C. However, at CRs of 5 C and 6 C, T_{SC} decreased significantly and the SC temperature decreased to 101.2 °C. This phenomenon is attributable to the rapid proliferation of Li dendrites within batteries subjected to high CRs, thus indicating a greater propensity for the occurrences of SCs and TRs. Fig. 10 (d) shows the SOC under an expansion-force rise rate of 5 N/s (SOC_{abn}), which ranged from 102.5 % to 111.3 % as the CR decreased. Notably, the SOC_{abn} for the batteries charged at a rate of 1 C was considerably higher than that of other batteries, whereas the SOC_{abn} for the other batteries was below

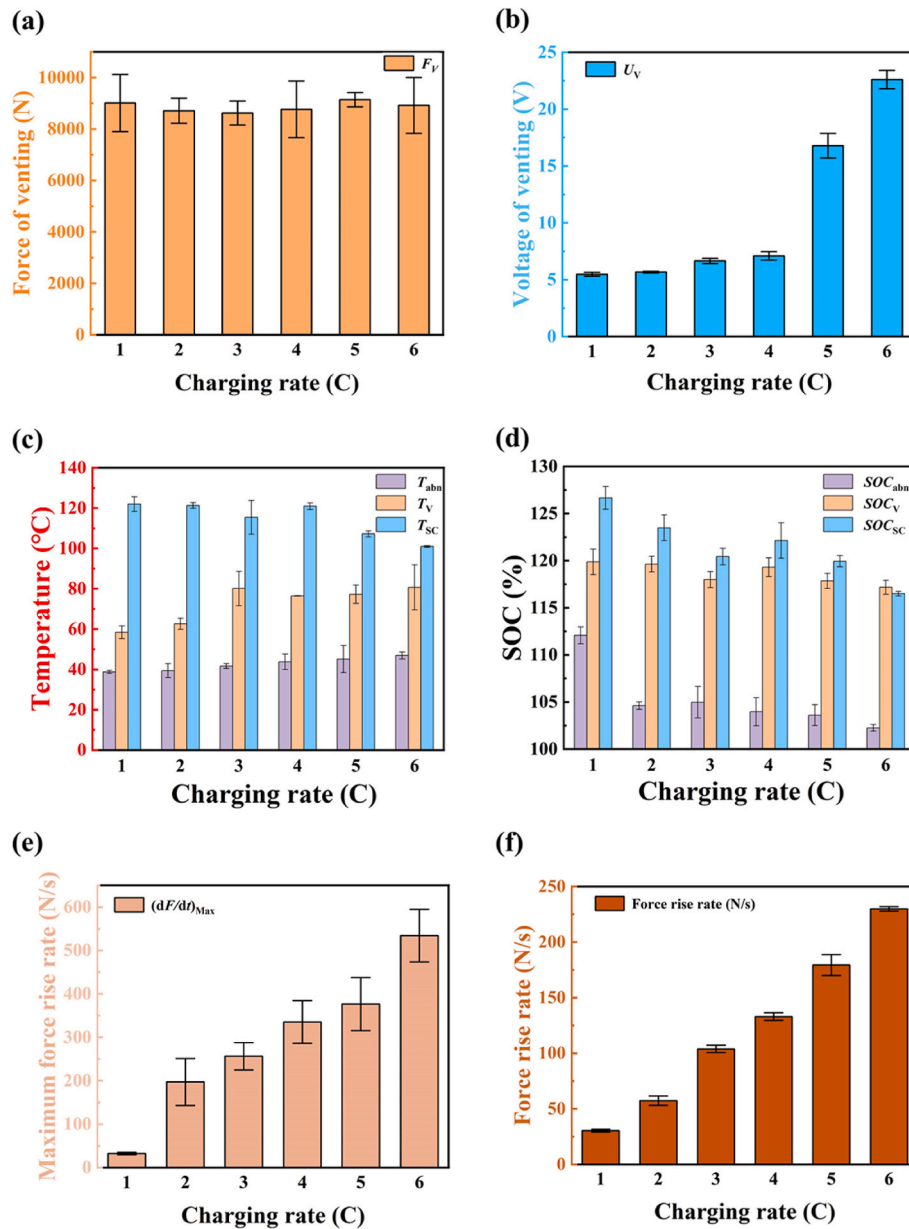


Fig. 10. Battery-safety boundary parameters. (a) Venting force; (b) venting voltage; (c) temperatures of $(dF/dt)_{abn}$, venting, and SC; (d) SOC of $(dF/dt)_{abn}$, venting, and SC; (e) maximum force rise rate; (f) linear fitting of force rise rate.

105 %. This suggests that by monitoring the expansion force anomaly during slight overcharging, an effective warning system for TR can be established. The SOC of venting (SOC_V) consistently exceeded 118 %. The SOC when the voltage drop rate reached its maximum (SOC_{SC}) showed a negative correlation with the CR, thus suggesting that a higher CR facilitates the occurrence of SCs. This is because, under high-rate rapid charging, the anode of the battery is more likely to undergo lithium plating, which can pierce the separator and result in SCs within the battery (Oka et al., 2023). Fig. 10 (e) shows the maximum force rise rate $((dF/dt)_{Max})$ before the safety valve is vented. Clearly, $(dF/dt)_{Max}$ increased continuously with the CR. In particular, the effect of the CR on $(dF/dt)_{Max}$ was extremely significant, with $(dF/dt)_{Max}$ reaching 31.4 N/s and 505.6 N/s at CRs of 1 C and 6 C, respectively. Fig. 10 (f) shows the expansion force rise rate and its linear fitting results during the plateau period, which is mentioned in Fig. 4. The fit approximated a straight line, and the R-square of the fitting curves was approximately 0.99. The fitted equation is shown in Eq. (11).

$$dF/dt = -17.92 + 39.53C \quad (11)$$

where dF/dt denotes the expansion-force rise rate in the expansion-force linear phase, and C refers to the CR.

Table 3 shows the comparison of abnormal time for multidimensional signals during battery TR under different CRs. Fig. 11 displays the moments of expansion force anomaly, SC, and TR during overcharging at various CRs. The definitions are as follows:

$$t_{abn} = t((dF/dt) = 5 \text{ N/s}) \quad (12)$$

$$t_{SC} = t((dU/dt) = \text{Max}) \quad (13)$$

$$t_{TR} = t((dT/dt) = \text{Max}) \quad (14)$$

As the CR increased, the signal warning moments exhibited a consistent trend of occurring progressively earlier. When the CR reaches 3 C, the time gap between the SC and TR is reduced significantly. Fig. 11 (b) depicts the time intervals when the expansion force signal exhibited abnormal behaviour prior to SC and TR, which are defined as follows:

$$\Delta t_1 = t_V - t_{abn} \quad (15)$$

$$\Delta t_2 = t_{SC} - t_{abn} \quad (16)$$

$$\Delta t_3 = t_{TR} - t_{abn} \quad (17)$$

As the CR increased, the effectiveness of the expansion force warning diminished. Quantitatively, the time interval between the warning signals of the expansion force and temperature increased steadily from 151 s to 682 s as the charging rate decreased from 6 C to 1 C. The force signal offered an early warning 682 s before the battery TR occurred at 1 C. However, even at a CR of 6 C, the expansion force-based TR warning still provided a warning time of 2.5 min. This provides sufficient time for individuals to implement precautionary actions to mitigate TR and protect human lives and property. In conclusion, the expansion-force anomaly moment based on the definition herein can provide an effective warning when overcharging occurs at a high CR.

This study delves into the critical matter of early safety alerts

Table 3

Comparison of abnormal time for multidimensional signals during battery TR under different charging rates.

Charging rate (C)	t_{abn} (s)	t_V (s)	t_{SC} (s)	t_{TR} (s)	Δt_1 (s)	Δt_2 (s)	Δt_3 (s)
1	4013	4274	4514	4695	261	501	682
2	1876	2146	2214	2429	270	338	553
3	1247	1426	1456	1494	179	209	247
4	927	1068	1089	1105	141	162	178
5	744	853	864	895	109	120	151
6	616	707	713	770	91	97	154

concerning the overcharging of ESSs. The proposed early warning approach aims to mitigate the risk of TR in batteries resulting from overcharging, which is anticipated to diminish the occurrence of safety incidents, as well as the environmental impact of pollutant emissions stemming from fires and explosions in ESSs. Additionally, it is expected to conserve energy by reducing the need for interventions like cooling and firefighting in the aftermath of safety accidents. The method introduced here is applicable across a broad spectrum of scenarios, yet it is not without its constraints. Specifically, its efficacy at the module and system levels requires further additional validation. For instance, in configurations involving multiple rows of modules, it becomes necessary to equip each row with a force sensor. Furthermore, the challenge of accurately identifying and localizing defective cells within a module in response to abnormal signals presents a significant area for future investigation.

4. Conclusions

In this paper, the thermal runaway process induced by overcharging is divided into five stages, and the variations in expansion force, gas, voltage, and temperature at different charging rates across each stage are comprehensively investigated. This study proposed an early warning approach based on the expansion force, which is proved to be highly effective under fast charging conditions. The primarily conclusions are as follows.

- (1) The expansion force first exhibited abnormal behaviours as compared with the temperature signal, which ranged only from 35.4 °C to 48.1 °C under charging rates of 1–6 C. The time interval between the expansion force and abnormal temperature moments increased from 151 to 682 s as the charging rate decreased. The expansion force provided a warning for battery TR by overcharging 11 min earlier, particularly at the 1 C charge condition.
- (2) Side reactions aggravate gas generation inside the battery as overcharging progresses. The battery was filled with gas, as detected via ultrasound, and the corresponding SOC decreased from 105 % to 103 % as the charging rate increased from 1 C to 6 C. However, the charging rate hardly affected the SOC boundary for venting, which remained at around ± 118 %.
- (3) When the charging rate was lower than 3 C, there would be a plateau stage was observed for the expansion-force rise rate during battery overcharging, while the plateau disappeared with charge rate higher than 3 C, and the expansion-force rise rate continued to increase. In addition, the maximum expansion-force rise rate at 6 C was 16.1 times higher than that at 1 C.

CRedit authorship contribution statement

Kuijie Li: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Long Chen:** Writing – original draft, Data curation. **Xuebing Han:** Writing – review & editing, Supervision, Resources. **Xinlei Gao:** Writing – review & editing, Investigation. **Yao Lu:** Writing – review & editing. **Depeng Wang:** Writing – review & editing. **Shun Tang:** Funding acquisition. **Weixin Zhang:** Writing – review & editing. **Weixiong Wu:** Writing – review & editing. **Yuan-cheng Cao:** Supervision, Resources. **Languang Lu:** Writing – review & editing, Supervision, Resources. **Jinyu Wen:** Supervision, Resources. **Shijie Cheng:** Supervision, Resources. **Minggao Ouyang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that may have influenced this study.

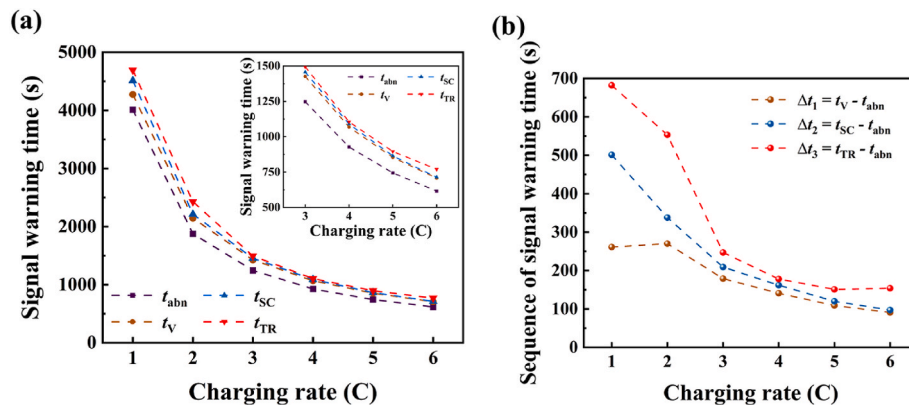


Fig. 11. Superiority of expansion force signal. (a) Signal warning moment; (b) sequence of signal-warning time interval.

Data availability

No data was used for the research described in the article.

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